

Famous Quote Quiz

Kotlin Multiplatform (KMP) TEST TASK

General information

In the Famous Quote Quiz game the system asks questions and the user tries to pick the correct answer. Depending on the selected mode, the user either chooses the correct answer from a list of answers, or simply answers Yes/No to the question ("Who said it?").

Task

Please create a tabbed cross-platform application with a splash screen and a login screen on application start. The client must be built with Kotlin Multiplatform and Compose Multiplatform. Quotes, users and quiz logic are served by a backend built with Spring Boot (Kotlin).

Technology stack

- ✓ **Client:** Kotlin Multiplatform (KMP) with Compose Multiplatform for the shared UI.
- ✓ **Backend:** Spring Boot (Kotlin) exposing a REST API, with its own database for quotes and users.
- ✓ The client talks to the backend over the API; the quote list must not be hard-coded in the app.

User Login Screen (no wireframe provided)

- ✓ On first start and after user logout the login screen should be shown.
- ✓ After a successful login the user should be redirected to the application main screen.
- ✓ The next time the user opens the application he should not be asked to login again unless he has previously logged out.
- ✓ The layout should contain input fields for email and password, error labels under each input field and button logic.
- ✓ Each input field should have a hint text. This text should match the field purpose.
- ✓ Error texts should be positioned under the respective input field. They should not be visible by default.
- ✓ There should be a simple validation for non-empty values.
- ✓ The validation error texts should be in red (**#f00**).
- ✓ The validation error views should be hidden when the user makes a new input.

Tab 1 – Main screen – Famous quote quiz screens

- ✓ A single quiz session should consist of 10 quotes. Changing the quiz mode should result in a session restart.
- ✓ Regardless of the current quiz mode, if the user picks the right answer, a modal box is displayed stating **"Correct! The right answer is: ..."**.
- ✓ If the user picks the wrong answer, the message should state **"Sorry, you are wrong! The right answer is: ..."**. Once the OK button is clicked, the next question should be shown.
- ✓ After the quiz session is completed, a statistics results screen with a **Start again** button should be shown (no wireframe provided).

Tab 2 – Settings Screen

- ✓ A single control switching between modes.
- ✓ Changing the mode should result in a quiz session restart.
- ✓ Binary (Yes/No) mode – the default mode.
- ✓ Multiple choice mode – showing three possible answers, only one is correct.

Tab 3 – User Profile Screen (no wireframe provided)

- ✓ Profile information should be displayed only. No editing required.
- ✓ The information will be based on the credentials filled in the login form.
- ✓ A logout button should be included. Logging out the user will redirect him to the login screen.

Backend – Spring Boot (Kotlin)

- ✓ Provide a REST API that serves the quotes and their authors, handles login/authentication, and exposes the profile information.
- ✓ Store quotes and users in a database (embedded is acceptable, e.g. H2). Seed a set of famous quotes on startup.
- ✓ The answer options for multiple choice mode (one correct + distractors) should be produced by the backend.
- ✓ Provide a short description of the API endpoints in the documentation.

Requirements

- ✓ Client – Kotlin Multiplatform with Compose Multiplatform. Primary target: Android 8.0 (API 26) and above. iOS (deployment target 15.0 and above) and Desktop (JVM 11 and above) targets are a welcome bonus but not required.
- ✓ Backend – Spring Boot (Kotlin) with a REST/JSON API and a database.
- ✓ Phone form factor.
- ✓ Both portrait and landscape orientations should be supported.
- ✓ Local persistence on the client (e.g. a local database for caching or the login session) is free – the candidate's choice. It is not mandatory.

Deliverables

- Source code (client and backend).
- Documentation (how to run the backend and the client, and a short overview of the API).
- Anything that candidate thinks that would be a good add-on.